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[3]: # =====
# QUERY 1: Trend Detection - Monthly Platform Engagement Tracking
# Logic: Groups posts by calendar month and platform to analyze temporal
#       trends in post volume and engagement distribution over time.
# =====

query_1 = """
    SELECT
        strftime('%Y-%m', timestamp) AS post_month,
        platform,
        COUNT(post_id) AS total_posts,
        SUM(likes) AS total_likes,
        ROUND(AVG(likes), 2) AS avg_likes_per_post
    FROM posts
    WHERE timestamp IS NOT NULL
    GROUP BY post_month, platform
    ORDER BY post_month DESC, total_likes DESC;
"""

result_1 = pd.read_sql_query(query_1, conn)
display(result_1)
```

post month	platform	total posts	total likes	avg likes per post
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[4]: # =====
# QUERY 2: Behavioural Grouping - Top Active Users by Engagement
# Logic: Aggregates total post count and combined interactions (likes, shares,
#       comments) per user to segment high-value active users from lurkers.
# =====

query_2 = """
    SELECT
        p.user_id,
        COUNT(p.post_id) AS total_posts_made,
        SUM(p.likes + p.shares + p.comments) AS cumulative_engagement,
        ROUND(AVG(p.likes), 2) AS avg_likes_per_post
    FROM posts p
    GROUP BY p.user_id
    ORDER BY cumulative_engagement DESC
    LIMIT 10;
"""

result_2 = pd.read_sql_query(query_2, conn)
display(result_2)
```

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[5]: # =====
# QUERY 3: Anomaly Discovery - Outlier Posts with Exceptional Engagement
# Logic: Identifies viral or anomalous posts where the like count significantly
#       exceeds the dataset average (using statistical thresholding).
# =====

query_3 = """
    SELECT
        post_id,
        user_id,
        platform,
        likes,
        shares,
        comments,
        timestamp
    FROM posts
    WHERE likes > (SELECT AVG(likes) + (3 * (SELECT AVG((likes - (SELECT AVG(likes) FROM pos
    ORDER BY likes DESC;
    """

result_3 = pd.read_sql_query(query_3, conn)
display(result_3)
```

post_id user_id platform likes shares comments timestamp